



SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS 8 / 8 completed

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error".

PROGRAM EDITOR

Java

Run

Save

```
1 public class main {
2     public static void main(String[] args) {
3         Logger logger = new Log4jLogger();
4         logger.debug("This is a debug message");
5         logger.info("This is an info message");
6         logger.warning("This is a warning message");
7         logger.error("This is an error message");
8     }
9 }
10 interface Logger {
11     void debug(String message);
12     void info(String message);
13     void warning(String message);
14     void error(String message);
15 }
16 class Log4jLogger implements Logger {
17     public void debug(String message) {
18         System.out.println("DEBUG: " + message);
19     }
20     public void info(String message) {
21         System.out.println("INFO: " + message);
22     }
23     public void warning(String message) {
24         System.out.println("WARNING: " + message);
25     }
26     public void error(String message) {
27         System.out.println("ERROR: " + message);
28     }
29 }
```

INPUT

Your INPUT goes here!



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the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error".

```
16 class Log4jLogger implements Logger {
17     public void debug(String message) {
18         System.out.println("DEBUG: " + message);
19     }
20     public void info(String message) {
21         System.out.println("INFO: " + message);
22     }
23     public void warning(String message) {
24         System.out.println("WARNING: " + message);
25     }
26     public void error(String message) {
27         System.out.println("ERROR: " + message);
28     }
29 }
```

OUTPUT

DEBUG: This is a debug message
INFO: This is an info message
WARNING: This is a warning message
ERROR: This is an error message



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QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement

PROGRAM EDITOR

Java

Run

Save

```
1 public class Main {
2     interface Character {
3         void move();
4         void attack();
5         void takeDamage(int damage);
6     }
7
8     static class Hero implements Character {
9         private int health;
10        public Hero(int health) {
11            this.health = health;
12        }
13        public void move() {
14            System.out.println("Hero is moving.");
15        }
16        public void attack() {
17            System.out.println("Hero is attacking.");
18        }
19        public void takeDamage(int damage) {
20            health -= damage;
21        }
22    }
23 }
```

INPUT

Your INPUT goes here!



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program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement

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```
hero.takeDamage(30);
villain.move();
villain.attack();
villain.takeDamage(40);
monster.move();
monster.attack();
monster.takeDamage(50);
}
```

OUTPUT

```
Hero is moving.
Hero is attacking.
Hero took 30 damage. Remaining health: 70
Villain is moving.
Villain is attacking.
Villain took 40 damage. Remaining health: 80
Monster is moving.
Monster is attacking.
Monster took 50 damage. Remaining health: 100
```



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Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         List<Integer> list = new ArrayList<>(Arrays.asList(5, 2, 9, 1, 6, 3));
5         Collections.sort(list);
6         System.out.println(list);
7     }
8 }
```

OUTPUT

[1, 2, 3, 5, 6, 9]

Your INPUT goes here!



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HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

```
1 import java.util.HashSet;
2 public class Main {
3     public static void main(String[] args) {
4         HashSet<String> set = new HashSet<>();
5         set.add("Apple");
6         set.add("Banana");
7         set.add("Mango");
8         set.add("Grapes");
9         String toRemove = "Banana";
10        set.remove(toRemove);
11        System.out.println("HashSet after removing " + toRemove + ": " + set);
12    }
13 }
```

OUTPUT

HashSet after removing Banana: [Apple, Grapes, Mango]

Your INPUT goes here!



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TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

```
1 import java.util.TreeMap;
2 import java.util.Map;
3 public class Main {
4     public static void main(String[] args) {
5         TreeMap<String, Integer> map = new TreeMap<>();
6         map.put("Apple", 10);
7         map.put("Banana", 20);
8         map.put("Cherry", 30);
9         for (Map.Entry<String, Integer> entry : map.entrySet()) {
10             System.out.println("Key: " + entry.getKey() + ", Value: " + entry.getValue());
11         }
12     }
13 }
```

OUTPUT

Key: Apple, Value: 10
Key: Banana, Value: 20

Your INPUT goes here!



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PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

```
1 import java.util.PriorityQueue;
2 import java.util.Scanner;
3 public class Main {
4     public static void main(String[] args) {
5         PriorityQueue<Integer> pq = new PriorityQueue<>();
6         Scanner sc = new Scanner(System.in);
7         if (sc.hasNextInt()) {
8             int n = sc.nextInt();
9             for (int i = 0; i < n; i++) {
10                 if (sc.hasNextInt()) {
11                     pq.add(sc.nextInt());
12                 }
13             }
14         }
15         System.out.println(pq);
16         sc.close();
17     }
18 }
```

OUTPUT

[]

Your INPUT goes here!

LinkedListHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
5         map.put("Apple", 10);
6         map.put("Banana", 20);
7         map.put("Cherry", 30);
8         map.put("Date", 40);
9         System.out.println("Keys in insertion order:");
10        for (String key : map.keySet()) {
11            System.out.println(key);
12        }
13    }
14 }
```

Your INPUT goes here!

OUTPUT

Keys in insertion order:

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

```
1 public class Main {
2     public static void main(String[] args) {
3         java.util.Scanner sc = new java.util.Scanner(System.in);
4         java.util.Set<Integer> set = new java.util.HashSet<>();
5         if (!sc.hasNextInt()) return;
6         int n = sc.nextInt();
7         for (int i = 0; i < n; i++) {
8             if (sc.hasNextInt()) {
9                 set.add(sc.nextInt());
10            }
11        }
12        if (!set.isEmpty()) {
13            int max = java.util.Collections.max(set);
14            int min = java.util.Collections.min(set);
15            System.out.println("Maximum value: " + max);
16            System.out.println("Minimum value: " + min);
17        }
18    }
19 }
```

Your INPUT goes here!

OUTPUT